AP CALCULUS AB

Unit 1 • Lesson 1.8 practice

Lesson 1.8 • Determining Limits Using the Squeeze Theorem

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x\to 0} x \sin\frac{1}{x}$ is

(A) 0

(B) 1

(C) It does not exist.

(D) ∞

Question 2

$\lim_{x\to 0} \frac{\sin 4x}{2x}$ is

(A) 4

(B) 2

(C) $\frac{1}{2}$

(D) 1

Question 3

$\lim_{x\to 0} \frac{1-\cos x}{x}$ is

(A) 1

(B) -1

(C) 0

(D) It does not exist.

Question 4

If $3 - x^2 \le f(x) \le 3 + x^2$ for all x, then $\lim_{x\to 0} f(x)$ is

(A) 0

(B) 6

(C) It cannot be determined.

(D) 3

Question 5 (Free response)

(a) Explain why $-|x| \le x\cos\frac{1}{x} \le |x|$ for $x \neq 0$, and use it to find $\lim_{x\to 0} x\cos\frac{1}{x}$.

(b) Find $\lim_{x\to 0} \frac{\tan x}{x}$.

(c) Find $\lim_{x\to 0} \frac{x}{\sin 3x}$.